

Grupa : 403
Specializarea : AiA
Nr. :
Data : 24.01.2021

EXAMEN LOGICĂ COMPUTATIONALĂ

① (a) $111110,001_{(2)} = ?_{(10)}$

$$111110,001_{(2)} = 1 \cdot 2^{-3} + 0 \cdot 2^{-2} + 0 \cdot 2^{-1} + 0 \cdot 2^0 + 1 \cdot 2^1 + 1 \cdot 2^2 + 1 \cdot 2^3 + 1 \cdot 2^4 + 1 \cdot 2^5$$
$$= \frac{1}{8} + 2 + 4 + 8 + 16 + 32 = 0,125 + 62 = 62,125_{(10)}$$

$$111110,001_{(2)} = 62,125_{(10)}$$

$$62,125_{(10)} = ?_{(2)}$$

$$62,125_{(10)} = 111110,001$$

$$62 : 2 = 31, \text{r}0$$

$$31 : 2 = 15, \text{r}1$$

$$15 : 2 = 7, \text{r}1$$

$$7 : 2 = 3, \text{r}1$$

$$3 : 2 = 1, \text{r}1$$

$$1 : 2 = 0, \text{r}1$$

$$0,125 \cdot 2 = 0,25 + 0 \Rightarrow a_{-1} = 0$$

$$0,25 \cdot 2 = 0,5 + 0 \Rightarrow a_{-2} = 0$$

$$0,5 \cdot 2 = 1 + 0 \Rightarrow a_{-3} = 1$$

(b) $+30 / -30$

$$30 : 2 = 15 \text{ r}0$$

$$15 : 2 = 7 \text{ r}1$$

$$7 : 2 = 3 \text{ r}1$$

$$3 : 2 = 1 \text{ r}1$$

$$1 : 2 = 0 \text{ r}1$$

$$\Rightarrow 30_{(10)} = 11110_{(2)}$$

- prin maxime și semn (cod direct)

$$+30_{(10)} = 00011110_{(2)}$$

$$-30_{(10)} = 10011110_{(2)}$$

- prin complement față de 2 (cod complementat)

$$+30_{(10)} = 00011110_{(2)}$$

↓

$$\begin{array}{r} 11100001 \\ 1+ \\ \hline 11100011_{(2)} = -30_{(10)} \end{array}$$

- prin complement față de 1 (cod invers)

$$+30_{(10)} = 00011110_{(2)}$$

$$-30_{(10)} = 11100001_{(2)}$$

② $x_1 \overline{x_2} + x_3 \overline{x_4} = (\overline{x_1} + x_2)(\overline{x_3} + x_4)$

x_1	x_2	x_3	x_4	$x_1 \overline{x_2}$	$x_3 \overline{x_4}$	$x_1 \overline{x_2} + x_3 \overline{x_4}$
1	1	1	1	0	0	1
1	1	1	0	0	1	0
1	1	0	1	0	0	1
1	1	0	0	0	0	1
1	0	1	1	1	0	0
1	0	1	0	1	1	0
1	0	0	1	1	0	0
1	0	0	0	1	0	0
0	1	1	1	0	0	1
0	1	1	0	0	1	0
0	1	0	1	0	0	1
0	1	0	0	0	0	1
0	0	1	1	0	0	1
0	0	1	0	0	1	0
0	0	0	1	0	0	1
0	0	0	0	0	0	1

x_1	x_2	x_3	x_4	$\overline{x_1} + x_2$	$\overline{x_3} + x_4$	$(\overline{x_1} + x_2)(\overline{x_3} + x_4)$
1	1	1	1	1	1	1
1	1	1	0	1	0	0
1	1	0	1	1	1	1
1	1	0	0	1	1	1
1	0	1	1	0	1	0
1	0	1	0	0	0	0
1	0	0	1	0	1	0
1	0	0	0	0	1	0
0	1	1	1	1	1	1
0	1	1	0	1	0	0
0	1	0	1	1	1	1
0	1	0	0	1	1	1
0	0	1	1	1	1	1
0	0	1	0	1	0	0
0	0	0	1	1	1	1
0	0	0	0	1	1	1

③ $\overline{x_2 x_3 x_4} + \overline{x_1 x_2 x_3}$

$$\overline{x_2 x_3 x_4} + \overline{x_1 x_2 x_3} = \overline{x_2 x_3 x_4} \cdot \overline{x_1 x_2 x_3} =$$

$$= (x_2 + \overline{x_3} + x_4) \overline{x_1} \overline{x_2} x_3 = \overline{x_1} x_3 + \overline{x_1} \overline{x_2} + \overline{x_1} \overline{x_2} x_3 x_4$$

④ $f(x_1, x_2, x_3) = x_2 \overline{x_3} + \overline{x_1} + x_1 x_2 \overline{x_3}$

FCD

$$f(x_1, x_2, x_3) = (x_1 + \overline{x_1}) x_2 \overline{x_3} + (x_2 + \overline{x_2}) \overline{x_1} + x_1 x_2 \overline{x_3}$$

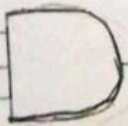
FCD

$$f(x_1, x_2, x_3) = x_1 x_2 \overline{x_3} + \overline{x_1} x_2 \overline{x_3} + (x_3 + \overline{x_3}) x_2 \overline{x_1} + (x_3 + \overline{x_3}) \overline{x_2} \overline{x_1}$$

FCD

$$f(x_1, x_2, x_3) = x_1 x_2 \overline{x_3} + \overline{x_1} x_2 \overline{x_3} + \overline{x_1} x_2 x_3 + \overline{x_1} \overline{x_2} \overline{x_3} + \overline{x_1} \overline{x_2} x_3$$

⑤ (a)

(1) $\begin{matrix} 1 \\ 0 \\ \overline{x} \end{matrix}$  $0 \Rightarrow 1 \cdot 0 \cdot \overline{x} = 0 \Rightarrow$

$\Rightarrow$ Valorile pe care le poate lua variabila x sunt: 0 SAU 1

a	b	c	f
1	1	1	0
1	1	0	1
1	0	1	0
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	0
0	0	0	0

*
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(2)



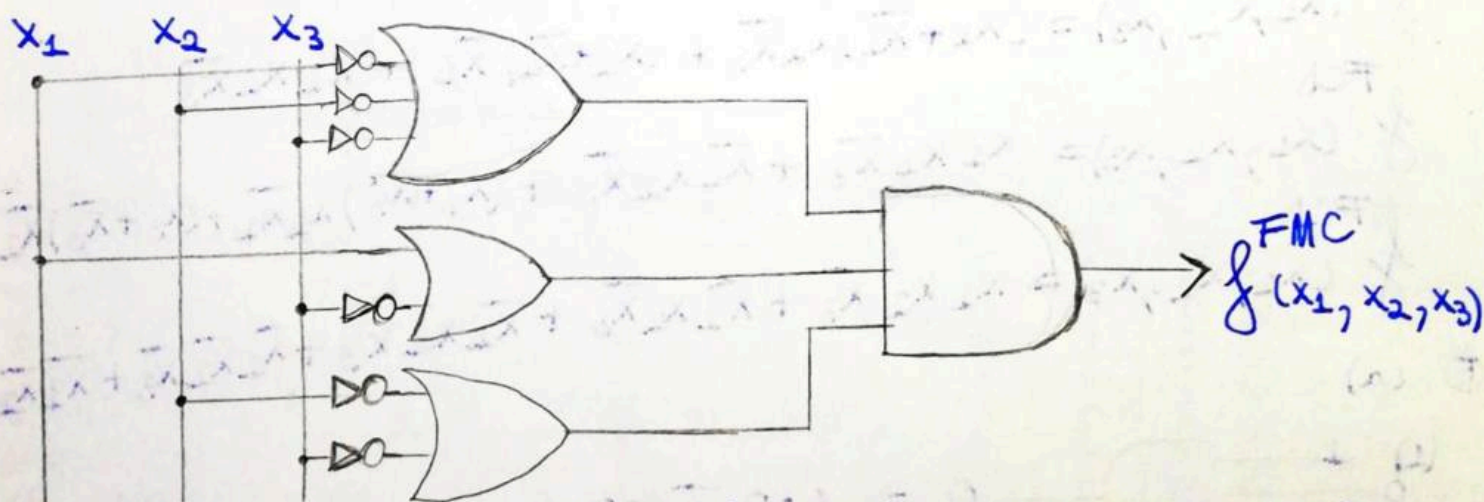
→ Valorile pe care le poate lua variabila x sunt: 0 SAU 1

a	b	f
1	1	1
1	0	0
0	1	0
0	0	1

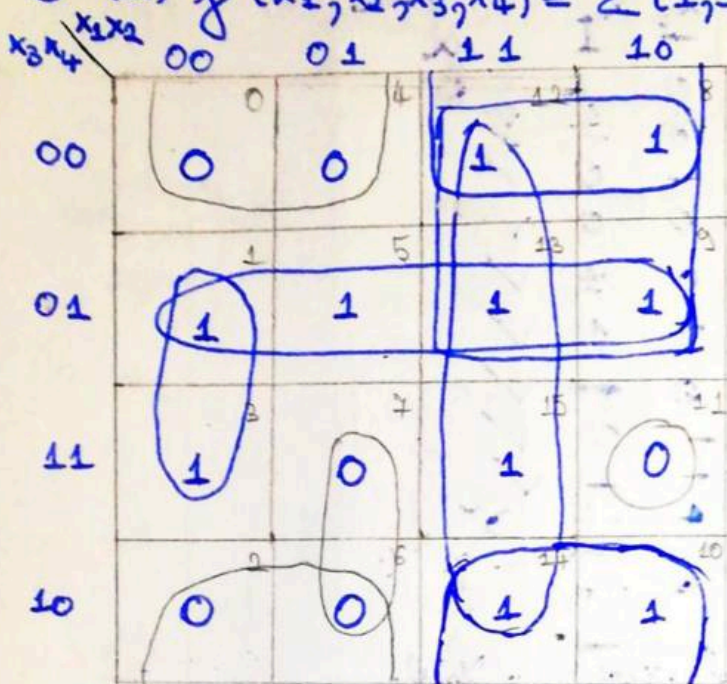
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(b) $f^{FMC}(x_1, x_2, x_3) = (\bar{x}_1 + \bar{x}_2 + \bar{x}_3)(x_1 + \bar{x}_3)(\bar{x}_2 + \bar{x}_3)$



6) (a) $f(x_1, x_2, x_3, x_4) = \sum (1, 3, 5, 8, 9, 10, 12, 13, 14, 15)$



FMD
 $f(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_2 x_4 + \bar{x}_3 x_4 + x_1 x_2 + x_1 \bar{x}_3 + x_1 \bar{x}_4$

$\bar{f}$ FMD
 $\bar{f}(x_1, x_2, x_3, x_4) = x_1 \bar{x}_4 + \bar{x}_1 x_2 x_3 + x_1 \bar{x}_2 x_3 x_4$

FMC
 $f(x_1, x_2, x_3, x_4) = (x_1 + x_4) \cdot (x_1 + \bar{x}_2 + \bar{x}_3) \cdot (\bar{x}_1 + x_2 + \bar{x}_3 + x_4)$

$\Rightarrow$ FMC
 $f(x_1, x_2, x_3, x_4) = (x_1 + x_4) \cdot (x_1 + \bar{x}_2 + \bar{x}_3 + x_4) \cdot (\bar{x}_1 + x_2 + \bar{x}_3 + x_4)$

$\Rightarrow$ FMC
 $f(x_1, x_2, x_3, x_4) = \bar{x}_2 + \bar{x}_3 + \bar{x}_4 + \bar{x}_1 \bar{x}_2 x_4 + \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_1 x_2 + x_1 + x_1 x_2 \bar{x}_3 + x_1 x_2 x_4 + x_1 x_2 \bar{x}_3 x_4 + x_1 \bar{x}_3 + x_1 \bar{x}_2 \bar{x}_3 + x_1 \bar{x}_3 + x_1 x_3 x_4 + \bar{x}_2 x_3 x_4 + \bar{x}_3 x_4 + x_1 \bar{x}_4 + x_1 \bar{x}_2 \bar{x}_4 + x_1 \bar{x}_3 \bar{x}_4 + x_1 + \bar{x}_2 + \bar{x}_3 \Rightarrow$

$\Rightarrow$ FMC
 $f(x_1, x_2, x_3, x_4) = x_1 +$

- 1 = 0001
- 3 = 0011
- 5 = 0101
- 8 = 1000
- 9 = 1001
- 10 = 1010
- 13 = 1101
- 12 = 1100
- 14 = 1110
- 15 = 1111

Grupă	Indicii tr. cuplate	Echivalentul linar				Impl. primi
		x_1	x_2	x_3	x_4	
0	1	0	0	0	1	✓
	8	1	0	0	0	✓
	9	0	0	1	1	✓
1	5	0	1	0	1	✓
	10	1	0	0	1	✓
	12	1	1	0	0	✓
2	13	1	1	0	1	✓
	14	1	1	1	0	✓
	15	1	1	1	1	✓

Grupă	Indicii ter. cuplați	Echivalentul binar				Impl. prime
		x_1	x_2	x_3	x_4	
0	1,3	0	0	1	1	B
	1,5	0	1	0	1	
	1,9	1	0	0	1	
	8,9	1	0	0	0	
	8,10	1	0	1	0	
1	8,12	1	1	0	0	✓
	5,13	1	1	0	1	
	9,13	1	1	0	1	
	12,13	1	1	0	1	
	12,14	1	1	1	0	
2	13,15	1	1	1	1	✓
	14,15	1	1	1	1	

Grupă	Indicii ter. cuplați	Echivalentul binar				Impl. prime
		x_1	x_2	x_3	x_4	
2,5,9,13	1,5,9,13	1	1	0	1	Δ
	8,12,9,13	1	1	0	1	
	8,10,12,14	1	1	1	0	
	12,14,13,15	1	1	1	1	
	1,9,5,13	1	1	0	1	

Termeni canonici		1	3	5	8	9	10	12	13	14	15
α	1,3	*	*								
β	1,5	*		*							
γ	14,15									*	*
Δ	1,5,8,13	*		*	*			*			
φ	8,12,9,13			*	*	*	*	*			
θ	8,10,12,14			*	*	*	*	*			
λ	12,14,13,15						*	*	*	*	*

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FMD

$$f(x_1, x_2, x_3, x_4) = \overline{x_1} \overline{x_2} x_4 + \overline{x_3} x_4 + x_1 x_2 + x_1 \overline{x_3} + x_1 \overline{x_4}$$

h) $f(x_1, x_2, x_3, x_4) = \sum (2, 6, 7, 8, 13, 15);$

$d = 3 = 5 = *$

x_3x_4	x_1x_2 00	01	11	10
00	0	0	0	1
01	0	d	1	0
11	d	1	1	0
10	1	1	0	0

FMD $= \bar{x}_1 x_3 + x_2 x_4 + x_1 \bar{x}_2 \bar{x}_3 \bar{x}_4$
 $f(x_1, x_2, x_3, x_4)$

$\bar{FMD}$
 $\bar{f}(x_1, x_2, x_3, x_4) = \bar{x}_1 \bar{x}_3 + \bar{x}_2 x_4 + x_1 x_2 \bar{x}_4$
 $+ x_1 x_3 x_4$

FMC
 $f(x_1, x_2, x_3, x_4) = (x_1 + x_3)(x_2 + \bar{x}_4)(\bar{x}_1 + \bar{x}_2 + x_4)(\bar{x}_1 + \bar{x}_3 + x_4)$

FMC
 $f(x_1, x_2, x_3, x_4) = (x_1 + x_3)(x_2 + \bar{x}_4)(\bar{x}_1 + \bar{x}_2 + x_4)(\bar{x}_1 + \bar{x}_3 + x_4)$